

Ensemble Learning with SVC, Random Forest, and XGBoost for Accurate Alzheimer's Disease Prediction

Namitha K Y

SoCSE, Presidency University,
Bangalore, India,

namitha.ky@gmail.com, namitha.2023cse0015@presidencyuni-
versity.in,

<https://orcid.org/0009-0009-4044-3395>

Srabana Pramanik

Assistant Professor-Senior Scale ,
SoCSE, Presidency University

Banagalore, India,

srabana.pramanik@presidencyuniversity.in,

ORCID Profile: <https://orcid.org/0000-0001-6108-631X>

Abstract—Alzheimer's Disease is still a progressive neurodegenerative condition and early detection is needed in order to effectively intervene. Conventional methods of diagnosis are plagued by accuracy, feature selection, and processing high-dimensional neuroimaging data. The aim of this research is to improve classification performance through the combination of SVC, K-Neighbors Classifier, Decision tree Classifier, Random Forest Classifier, XGBClassifier optimization with Machine Learning. The method is by using search strategies to improve the diversity and stability of the hybrid model that guarantees balanced precision, F1-Score and recall. The ensemble model with its methods proposed here performs better than standard methods in the accuracy, sensitivity, and specificity of classification, ensuring it can minimize false positives and false negatives. Experimentation proves the model's superiority in working with complex medical data, affirming its prospects for automated AD diagnosis. The optimized classification setup offers an encouraging solution for improving computational methods for neurodegenerative disease detection.

Keywords: SVC, KNeighbors Classifier, Decision Tree Classifier, Random Forest Classifier, XGB Classifier, Ensemble model

I. INTRODUCTION

AD is an incurable neurodegenerative disorder with signs and symptoms of cognitive impairment, memory loss, and behavioral changes. It mainly occurs in elderly individuals, who gradually lose their ability to function in daily life. The disorder is caused by abnormal depositions of proteins in the brain in the form of beta-amyloid plaques and tau tangles, leading to neuronal loss and dysfunction. Gradually, the patient undergoes from severe cognitive impairment, faulty reasoning, and difficulty with communication. Brain atrophy increases with the advancement of the disease, leading to severe volume loss in gray matter and disrupting neural connections. Early detection and treatment are crucial in keeping the advancement of the disease under control and improving patient outcomes. AD is not only limited to cognitive impairment and affects individuals as well as their families. Patients usually experience psychological distress, and consequently leads to mood swings, depression, and increased dependence on caregivers. Its chronic nature imposes emotional and economic burdens on families that require long-term care arrangements. With constant degeneration of the neurons, the patients also experience difficulty in speech, problem-solving capacities, and the capacity to perform motor functions. Early-stage

classification of AD with the use of medical and demographic information related to Alzheimer's disease using ensemble model. The setting is restricted OASIS dataset, with the expectation of and correct clinical annotation. The model proposed in this work contain demographic information. The deep learning model like SVC, K-Neighbors classifier, Decision tree classifier, Random forest classifier, XGBClassifier, Ensemble model are used to predict early diagnosis of the disease. The ensemble model with Gradient Boosting Classifier gives the accuracy of 0.95 by combing all the model.

II. RELATED WORK

The study on ML introduces a designed system for medical image order and recognition of the Alzheimer's complaint. Looking to as Neuro-imaging grounded Beforehand Discovery of Alzheimer's complaint using Deep literacy (our proposed scheme) this paper gives a computer-backed individual(CAD) system aimed precisely grading stages of Alzheimer's complaint enhancing individual accuracy. The proposed model ResNet-50 with Alex Net armature using resembling processing capabilities of GPU employing the CUDA programming system which allows to exploit resembling processing capabilities NVIDIA GPUs. Neural network operations into parallelizable task similar as matrix proliferations and complications which executed coincidentally in multiple GPU cores. The community achieved launching vestments and organized to blocks and grids perform calculations on different subsets of the input data coincidentally. Different performance criteria including delicacy, perceptivity(SEN), F1-score particularity(SPC) employed to estimate the proposed scheme model with different classifiers similar as ReLU, Softmax, Sigmoid, SVM, RF attained loftiest delicacy 99.57[1].

The first proposed methodology XGBoost and ANN networks given extract appearance. MobileNet, Densnet, Googlenet models prototype. The following anticipated methodology is based on XGBoost and ANN with topographies on dissimilar models like Mobile Net, DenseNet121, DenseNet121, Google Net, MobileNet, Google Net, XG Boost and ANN profession.

Third future methodology is founded on Boost and ANN networks with plan different CNN prototype and handcrafted outline gives dissimilar MobileNet, Densenet121-handcrafted, DenseNet121-GoogLeNet-handcrafted MobileNet, Google Net handcrafted. The proposed given XGBoost, ANN networks gives different multiple handcrafted

CNN outline achieved better result with good technology. The MobileNet, GoogleNet-handcrafted XGboost prototype gives an curve 98.82% with accuracy 98.8%, sensitivity 98.9%, precision 97.08%, and specificity 99.5%[2].

The work ensemble representative Logistic Random Forest Boosting (LRFB) expressive ensemble of “Logistic Regression” (LR), “Random Forest” (RF) and “Gradient Boost” (GB). The future LRFB beat LR, RF, GB, “k-Nearest Neighbor” (k-NN), “Multi-Layer Perceptron” (MLP), “Support Vector Machine” (SVM), “AdaBoost” (AB), “Naïve Bayes” (NB), “Boost” (XGB), Decision Tree” (DT) and other ensemble ML copies with deference to the presentation metrics “Accuracy”, “Recall”, “Precision”, and “F1-Score” (FS). LRFB model outdone additional ML replicas via 55.18%, 52.88%, 50.54%, and 60.06%, correspondingly [4].

This work presents a novel attention-based multimodal feature fusion module Alzheimer's disease classification. We determined the best techniques for precise categorization by utilizing the ADNI dataset and contrasting different pre-trained models. Before feature fusion, the attention mechanism was incorporated, which greatly enhanced performance and produced high accuracy throughout several classification stages[5].

This exploration paper gives a different method by joining DenseNet-121-point birth by MRI images followed by Support Vector Machine(SVM) bracket unique different level of Alzheimer's disease. DenseNet-121 prototype honored its connection and well point exercise to apply to prize high-position, discrimination features from MRI reviews. The extracted features are to train the SVM bracket fine-tuned to get optimal effectiveness. Prototype has been estimated ADNI that contains MRI images labeled colorful level of Alzheimer's complaint. Its delicacy 92.5, perfection 91.2%, recall of 94.1% and F1-score of 92.6%[6].

III. METHODS

This paper presents ML model for early prediction of AD from the demographic data set of Alzheimer patients. The data set contain 2149 records with features like functional assessments, ADL, MMSE, Diet Quality etc. The data set is subjected to preprocessing techniques solely to retain numerical values. Data set is split into Training set (1719, 32) and Testing set shape (430, 32).

A. Preprocessing methods

Information and Missing Values Check: It outputs the compact information of the Data Frame with data types and non-null counts for every column through `df.info()` function.

It further identifies and prints the missing values in each column using `df.isnull().sum()` functions.

Handle Categorical Features: It finds columns having 'object' data type (categorical columns) and prints their counts. It then shows how to handle the categorical column by converting it into numerical representation using Label Encoder.

Deal with Numerical Features: It detects numerical columns and shows how to apply Min_Max Scaler (or Standard Scaler) to scale numerical features (in this case, 'Age' and 'MMSE') to a particular range (0 to 1 for MinMaxScaler).

Handle Missing Values: It loops over all columns and fills missing values. For object type columns, it replaces missing

values with the mode (most frequent). For numerical columns, it replaces missing values with the mean. Model performs feature selection using three different methods.

B. Feature selection

Correlation Analysis: It computes the numerical feature correlation matrix and plots it as a heat map. It proceeds to detect and recommend dropping highly correlated features (having a correlation coefficient greater than 0.9) to prevent multicollinearity problems.

SelectKBest (ANOVA F-value):

This uses the SelectKBest function with the ANOVA F-value as the score metric to rank features according to their statistical significance against the target variable ('Diagnosis'). It prints the scores for all the features and picks the top 10 features with the highest scores.

Recursive Feature Elimination (RFE) using Logistic Regression:

It uses RFE with a Logistic Regression model to remove features recursively and choose the top 10 features that most contribute to the performance of the model. The top 10 features that contribute most to the model's performance. The Fig.1 shows top 10 features are extracted from data set using random forest model. The extract features contain the better precision for compare with other data set values. Functional assessment, ADL, MMSE have significant role in identifying AD.

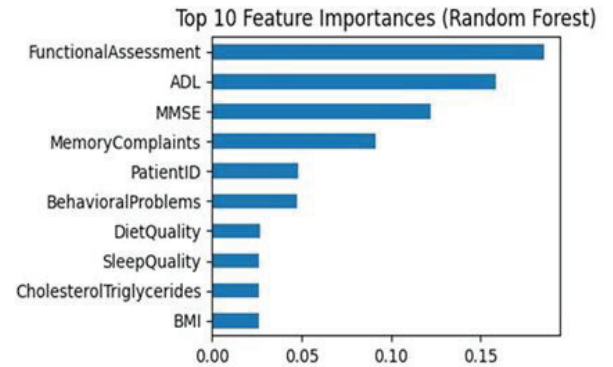


Fig. 1. Top 10 selected features from Random Forest

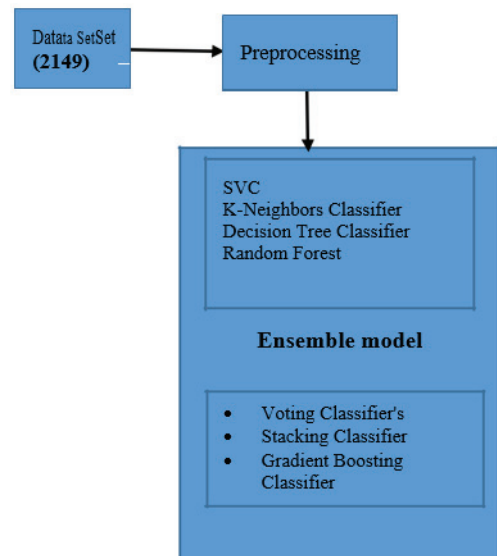


Fig. 2. Block Diagram of Machine Learning Model

The Figure. 2 show the Block diagram of the model ,2149 data set has been processed using preprocessing techniques. Missing values are eliminated, Handle Categorical Features, Deal with Numerical Features. After data preprocessing ML model has been trained with test and training data.

SVC

SVC (Support Vector Classification) algorithm is a supervised machine learning ideal secondhand for arrangement tasks to discover the best hyperplane that divides facts of different sessions with the all-out boundary. Since Diagnosis' is a binary variable (0 or 1), classification models are appropriate to provides accuracy of 0.65. For class 0 the Precision is 65% of predictions for class 0. Recall is 1.00 for all actual class 0 instances (278) were correctly identified. The predicted F1-score is 0.79 which gives good balance due to perfect recall and moderate precision. The Support identify 278 that were actual instances of class 0. The Class 1 has Precision of 0.00, model didn't correctly predict class 1 at all. Recall is 0.00 none of the 152 class 1 samples were correctly identified. F1-score is 0.00 gives worst possible performance. Support is 152 There were 152 actual instances of class 1.

K-Neighbors Classifier

K-Neighbors Classifier is a small and very popular supervised machine knowledge process castoff that makes use of distance metric to classify values. It provides accuracy of 0.57.

Decision Tree Classifier

Decision Tree Classifier is a supervised machine learning classifier to perform classification. It builds a tree structure of decisions as a function of the input feature to forecast the target class. It takes the parameter max_depth, min_samples_spilt, min_samples_leaf, random state. It provides the accuracy of 0.89.

Random Forest Classifier

Random Forest Classifier is a strong ensemble learning model that is applied to classification (and regression) problems. It creates an array of decision trees and predicts the class from the mean value of the predictions. estimators, max_depth, min_samples_split, random state as the highest priority for feature extraction. It provides accuracy of 0.93.

XGB

The XGB is an efficient gradient boosting algorithm with high performance that performs classification and regression. Applies gradient descent to reduce a loss function. Derives feature_learning_rate, subsample, colsample_bytree, objective, eval_metric, produce accuracy of 0.94.

Ensemble Model

An Ensemble Model is a machine learning algorithm that takes several base models and generates a more robust and stronger model. The concept is that several weak learners (models of average accuracy) can collectively provide better accuracy than one model. It employs the Voting, Stacking, boosting technique where it aggregates outputs from several models by majority voting for classification.

IV. RESULTS AND DISCUSSIONS

This section points out the outcomes of the methodology proposed being used to come up with early detection for AD

patients using a deep learning technique. The method was derived from the use of SVC, K-Neighbors Classifier, Decision Tree Classifier, Random Forest Classifier, XGB, Ensemble Model. The accuracy graph of combine machine learning model is represented by figure 3 where XGB provides the best accuracy 0.94. The Table 1 provides the classification of svc model with precision, recall F1-Score with accuracy of 0.65.

TABLE I. CLASSIFICATION OF SVC

Metric	Class 0	Class 1	Accuracy	Macro Average	Weighted Average
Precision	0.65	0.00	0.65	0.3	0.42
Recall	1.00	0.00	-	0.50	0.65
F1-Score	0.79	0.00	-	0.39	0.51
Support (Count)	278	152	430	430	430

Table 2 provides the Group of K- Neighbors Classifier with delicacy of 0.57. Decision Tree Classifier with delicacy of 0.89 for class 0, class 1. Random Forest with delicacy of 0.93, XGB with delicacy of 0.94 presented in the tables are reported in the paper.

TABLE II. CLASSIFICATION OF K -NEIGHBORS CLASSIFIER

Metric	Class 0	Class 1	Accuracy	Macro Average	Weighted Average
Precision	0.65	0.37	0.57	0.51	0.55
Recall	0.73	0.28	-	0.51	0.57
F1-Score	0.69	0.32	-	0.51	0.56

TABLE III. CLASSIFICATION OF DECISION TREE CLASSIFIER

Metric	Class 0	Class 1	Accuracy	Macro Average	Weighted Average
Precision	0.92	0.82	0.89	0.87	0.89
Recall	0.90	0.86	-	0.88	0.89
F1-Score	0.91	0.84	-	0.88	0.89
Support (Count)	278	152	430	430	430

TABLE IV. CLASSIFICATION OF RANDOM FOREST

Metric	Class 0	Class 1	Accuracy	Macro Average	Weighted Average
Precision	0.93	0.94	0.93	0.94	0.94
Recall	0.97	0.87	-	0.92	0.93
F1-Score	0.95	0.90	-	0.93	0.93
Support (Count)	278	152	430	430	430

TABLE V. CLASSIFICATION OF XGB

Metric	Class 0	Class 1	Accuracy	Macro Average	Weighted Average
Precision	0.94	0.94	0.94	0.94	0.94
Recall	0.97	0.89	-	0.93	0.94
F1-Score	0.96	0.92	-	0.94	0.94
Support (Count)	278	152	430	430	430

TABLE VI. ACCURACY OF VOTING CLASSIFIER'S

Class	Precision	Recall	F1-Score	Support
0	0.94	0.97	0.96	278
1	0.94	0.89	0.92	152
Accuracy	-	-	0.94	430
Macro Average	0.94	0.93	0.94	430
Weighted Average	0.94	0.94	0.94	430

TABLE VII. ACCURACY OF STACKING CLASSIFIER

Class	Precision	Recall	F1-Score	Support
0	0.94	0.97	0.96	278
1	0.94	0.89	0.92	152
Accuracy	0.94	-	-	430
Macro Average	0.94	0.93	0.94	430
Weighted Average	0.94	0.94	0.94	430

TABLE VIII. ACCURACY OF GRADIENT BOOSTING CLASSIFIER

Class	Precision	Recall	F1-Score	Support
0	0.96	0.97	0.96	278
1	0.93	0.89	0.92	152
Accuracy	-	-	0.95	430
Macro Average	0.94	0.93	0.94	430
Weighted Average	0.95	0.94	0.94	430

Ensemble model that combines all the model to elect the stylish among all using voting, Stacking, grade Boosting. The training data(X_train, y_train) is used to train the voting classifier, prognostications are done on the test data(X_test) with delicacy of 0.94. mounding classifier defines that a Logistic Retrogression model will act as a meta- learner to consolidate the prognostications of the base estimators with delicacy of 0.94. grade Boosting Classifier with 0.95 accuracy.

The Figure 4 is representing the delicacy of all the system used in ensemble model where grade Boosting give the maximum delicacy of 0.95 in prognosticating Alzheimer's complaint for the given data set.

The Figure 3 and Figure 4 are helpful for visualizing the training process and detecting possible problems such as overfitting or under fitting. The first plot displays how the accuracy of the model varied on the training and test sets throughout the training. You can observe whether the model is getting better with time and whether there is a big gap in training and test accuracy, which might be an indication of overfitting. The second plot illustrates how the loss of the model reduced on both test and training sets during training. Loss decreasing implies that the model is learning and improving its predictions.

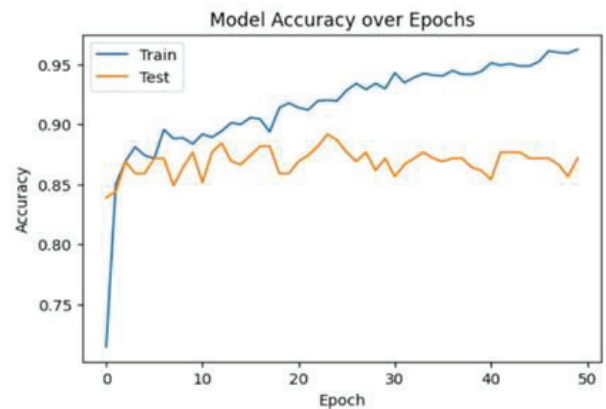


Fig. 3. Model's accuracy changed on training and test sets

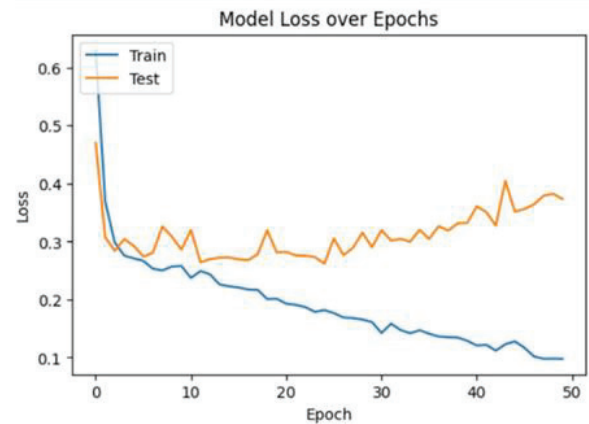


Fig. 4. Model loss decreased on both training and test sets

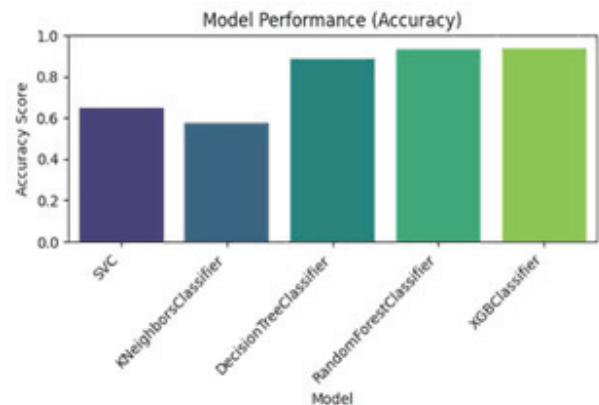


Fig. 5. Comparision of Model accuracies

Based on the analysis of Figure 5 represents the Deep Learning model achieved the highest accuracy, followed by the Support Vector Machine (SVC) and then the Decision Tree. Here are the specific accuracy scores of Deep Learning KNeighborsClassifier (0.57), Decision Tree Classifier (0.89), Random Forest Classifier (0.93), XGBClassifier(0.94).

V. CONCLUSION AND FUTURE WORK

These are very promising findings, showing the great potential of the current deep learning models for the early detection and staging of Alzheimer's disease based on MRI brain scans. Transfer learning enables the model to retain the knowledge extracted from other domains and thus be more efficient in training and performances. Alzheimer's complaint is a largely serious treatment for the case in the original stages. The original stage identifies advertisement to reduce the impact of advertisement on degeneration of neuron. We also employed machine knowledge algorithms to predict Alzheimer's complaint. Alzheimer's complaint categorizes use crossbred model to determine delicacy and delicacy graph. SVC(0.65), KNeighbors Classifier (0.57), DecisionTreeClassifier (0.89), RandomForest Classifier(0.93), XGBClassifier(0.94). Ensemble model with grade Boosting Classifier with 0.95 delicacy and can be attained. The future compass is MRI data set to determine the early prophecy of advertisement Disease. The Limitation of the model is clinical data, MRI, FMIR, PET scan images can be considered for processing and predicting the AD.

REFERENCES

- [1] N. N. Albogami, "early-stage detection and classification of Alzheimer's disease using cutting edge technologies," *j. theory. appl. inf. technol.*, vol. 103, no. 5, 2025.
- [2] A. M. Alayba, E. M. Senan, and J. S. Alshudukhi, "Enhancing early detection of Alzheimer's disease through hybrid models based on feature fusion of multi-CNN and handcrafted features," *Sci. Rep.*, vol. 14, no. 1, p. 31203, 2024.
- [3] K. Thanagraj, S. Kumar, S. Arul, K. Subramaniam, and W. Khairunizam, "Classification of Alzheimer using machine learning algorithms," in *AIP Conference Proceedings*, AIP Publishing LLC, 2025, p. 030013.
- [4] K. A. Shastri and S. A. Sattar, "Logistic random forest boosting technique for Alzheimer's diagnosis," *Int. J. Inf. Technol.*, vol. 15, no. 3, pp. 1719–1731, Mar. 2023, doi: 10.1007/s41870-023-01187-w.
- [5] A. M. Elassy, H. Amer, H. M. Ibrahim, and M. A. Mohamed, "Elevating Alzheimer's Diagnosis based on Attention-Guided MRI Feature Fusion," *Int. J. Telecommun.*, vol. 5, no. 01, pp. 1–20, 2025.
- [6] F. Hossain, R. K. Halder, and M. N. Uddin, "An integrated machine learning based adaptive error minimization framework for Alzheimer's stage identification," *Intell.-Based Med.*, vol. 11, p. 100243, 2025.